

Headline Clickbait Detection

A Multi-Approach NLP Study

Vishnu IMT2023097 Abhijit IMT2023054
Praneeth IMT2023555 Adwait IMT2023502

April 23, 2026

Abstract

This report investigates headline-level clickbait detection as a binary classification problem using three complementary NLP paradigms: *Lexical*, *Syntactic*, and *Semantic*. Each approach is developed independently and then combined in a Feature Fusion model. Evaluated on the Webis Clickbait Corpus (2017 train/test) and cross-dataset on the 2016 corpus, the lexical system achieves **85.2%** accuracy and **69.9% F1** on in-distribution data (validation F1: **74.6%**, a +11.6% gain over baseline). The semantic approach, using transformer-based sentence embeddings with optimised SVM classification (`paraphrase-mpnet-base-v2`, 768-d vectors), achieves **85%** test accuracy after resolving a 3.2-to-1 class imbalance via balanced weighting and RBF kernel tuning through GridSearchCV.

1 Introduction

Clickbait refers to online headlines engineered to attract clicks through curiosity gaps, emotional triggers, or misleading framing rather than genuine informational content. As news consumption increasingly shifts to social media, automated clickbait detection has become critical for computational journalism and NLP.

This project investigates clickbait detection at the **headline level**, framed as a binary classification task ($y \in \{0,1\}$). We adopt a *feature-engineering paradigm* where linguistic signals are extracted at multiple granularities—lexical, syntactic, and semantic—and fed to classical classifiers. This design enables:

- Full interpretability via classifier coefficients;
- Systematic ablation of each feature group;
- Modular fusion of all three approaches.

The remainder of this report is structured as follows: §2 formalises the problem, §3 details each of the three NLP approaches, §4 describes the combined

feature fusion model, §5 presents unified results, §6 discusses key findings, and §7 concludes.

2 Problem Statement

Task. Given a news headline h , predict label $y \in \{0,1\}$ where $y=1$ indicates clickbait. The challenge involves three axes:

1. **Stylistic subtlety** — capturing curiosity-gap phrasing, emotional triggers, and second-person framing.
2. **Cross-dataset robustness** — generalising across corpora from different years as clickbait language evolves.
3. **Interpretability** — providing linguistically meaningful explanations of model decisions.

Datasets.

Table 1: Dataset Overview

Dataset	Samples	Role	Type
Webis-CB-17 train	≈2,500	Training	In-dist.
Webis-CB-17 valid	≈19,500	Validation	In-dist.
Combined train	21,926	Training	In-dist.
Webis-CB-17 test	18,979	Evaluation	In-dist.
Webis-CB-16 test	2,992	Evaluation	Cross-dataset

Training uses an **80/20 stratified split** (17,541 train / 4,385 validation). All vectorisers are fitted exclusively on the training partition to prevent data leakage.

3 Methodology

3.1 Lexical Approach

The lexical approach models headlines through surface-form text statistics, constructing a rich feature space from word distributions, character patterns, and hand-engineered domain signals—without syntactic parsing or semantic embeddings.

3.1.1 Feature Engineering Pipeline

(a) **Word TF-IDF N-grams.** A TF-IDF vectoriser with n -gram range (1, 3) forms the backbone of the feature space. Key design choices:

- *Sublinear TF scaling* ($\log(1+tf)$): dampens repeated clickbait trigger words (e.g., *SHOCKING SHOCKING*).
- *Trigrams*: capture complete clickbait idioms such as “*you won’t believe*” or “*what happens next*” that bigrams split across tokens.
- 10,000 features with `min_df=2`, `max_df=0.9` to filter noise while retaining distinctive vocabulary.

(b) **Character N-grams.** A secondary TF-IDF vectoriser (character level, range (3, 5), 5,000 features) captures:

- Punctuation styles: patterns like `!!!`, `???` are strong signals.
- Morphological cues: prefixes and suffixes are vocabulary-agnostic.
- Cross-dataset generalisation: valuable when vocabulary drifts between corpora.

(c) **Handcrafted Domain Features (19 total).** Three categories of hand-engineered features are appended:

- *Length* (2): character count, word count.
- *Punctuation* (5): presence/count of `?` and `!`, uppercase word count.
- *Content* (12): second-person pronouns, clickbait phrase lexicon (10 patterns), number-leading headlines, question-word starts, digit presence, average word length, superlatives, emotional words.

Standalone F1 is modest (63.2%), but these features contribute interpretability and encode domain priors that TF-IDF alone cannot represent.

(d) **Feature Selection.** Chi-squared selection (`SelectKBest`, $k=8,000$) reduces the 15,019-feature space by **46.7%** with no loss in validation F1, removing noise and improving inference speed.

3.1.2 Classifier & Configuration

Logistic Regression is selected after a four-way comparison (Table 2). It outperforms alternatives while providing probability outputs essential for threshold tuning.

Final hyperparameters: $C=0.5$ (stronger regularisation via grid search), `class_weight=‘balanced’`, decision threshold 0.55 (validation) / 0.60 (2017 test).

3.1.3 Ablation Study

Eleven progressive experiments trace the full optimisation trajectory (Table 3).

Table 2: Classifier Comparison (Word TF-IDF, fixed features)

Classifier	Val F1 (%)
Logistic Regression	66.0
Linear SVM	65.8
SGD Classifier	65.2
Naïve Bayes	63.5

Table 3: F1 Progression Across 11 Experiments (Validation Set)

Exp.	Modification	Val F1 (%)	Δ
1	Baseline: TF-IDF (1, 2), LR, $C=1.0$	63.0	—
2	N-gram range (1, 3)	66.0	+3.0
3	Classifier selection (LR retained)	66.0	+0.0
4	<code>class_weight=‘balanced’</code>	66.0	+0.0
5	Sublinear TF, 10K feat., <code>max_df=0.9</code>	67.0	+1.0
6	+ Character n-grams (3, 5)	67.0	+0.0†
7	+ 19 handcrafted features	67.0	+0.0†
8	Chi-sq. selection ($k=8,000$)	67.0	+0.0
9	Tuned $C=0.5$	68.0	+1.0
10	Threshold optimisation (0.55)	69.0	+1.0
11	Full pipeline (final)	74.6	+5.6
Total improvement			+11.6

† Char and handcrafted features show additive gains when combined with later tuning; isolated contribution is 0% here but essential for the Exp. 11 jump.

3.2 Syntactic Approach

The syntactic approach analyses *how* a headline is written grammatically, deliberately ignoring vocabulary. Clickbait headlines tend toward simple, pronoun-led, engagement-driven structures, while non-clickbait is more formal and information-dense. Consider:

- **Clickbait:** “*You won’t believe this*” → PRON AUX VERB DET
- **Non-clickbait:** “*Government approves new policy*” → NOUN VERB ADJ NOUN

Even without words, the grammatical structure differs markedly. Three progressively richer syntactic models are developed.

3.2.1 Model 1: Handcrafted Syntactic Features

Explicit grammatical features are extracted per headline covering three groups:

- *POS tag counts*: pronouns, verbs, determiners, adjectives, etc.
- *Dependency relation counts*: subject→verb patterns and similar.
- *Structural cues*: `starts_with_pronoun`,

num_tokens, is_question, num_exclam.
Example: “You need to see this!” $\Rightarrow$ pos_PRON=1,
 pos_VERB=2, starts_with_pronoun=1,
 num_exclam=1

Classifier: Linear SVM. The handcrafted features produce linearly separable patterns: signals such as `starts_with_pronoun=1` or `num_exclam>0` point cleanly toward one class, making a linear decision boundary both sufficient and interpretable.

3.2.2 Model 2: POS Sequence N-grams (POS-TF-IDF)

Rather than counts, this model captures *ordering* of POS tags via TF-IDF over bigram and trigram POS sequences.

Example: “You won’t believe this” $\rightarrow$ PRON AUX VERB DET $\rightarrow$ TF-IDF learns that PRON AUX VERB is a clickbait-indicative trigram.

Classifier: Linear SVM. POS n-gram TF-IDF yields high-dimensional sparse features for which SVM is a well-established, strong choice.

3.2.3 Model 3: Combined (Handcrafted + POS-TF-IDF)

The combined model concatenates both feature sets to exploit complementary signals: count-level structure from Model 1 and sequential POS patterns from Model 2.

3.2.4 Syntactic Model Comparison

Table 4: Syntactic Approach: Model Comparison Across Datasets

Model	2017 Test		2016 dataset	
	Acc	F1	Acc	F1
Handcrafted features	0.79	0.80	0.78	0.76
POS-TF-IDF	0.79	0.81	0.79	0.79
Combined (M1 + M2)	0.80	0.81	0.79	0.77

Key observations:

- Clickbait consistently uses pronoun-led openings (*You, This*) and simple verb structures—captured cleanly by M1.
- POS n-grams (M2) capture stylistic sequence patterns and generalise better cross-dataset (2016 F1: 0.79 vs 0.76 for M1).
- Combining features (M3) yields the highest in-distribution accuracy (0.80) but does not uniformly improve cross-dataset F1, suggesting M1’s count features introduce some vocabulary-adjacent brittleness.

3.3 Semantic Approach

The semantic approach encodes headlines as dense contextual vectors using pre-trained Sentence Transformers, then trains discriminative classifiers on those representations. Unlike the lexical approach, no hand-crafted features are required; semantic meaning is captured implicitly by the encoder.

3.3.1 Phase 1 — Baseline Embeddings

An initial pipeline used `all-MiniLM-L6-v2` (384-d vectors) as a lightweight encoder. Three un-tuned classifiers were evaluated: SVM (linear kernel), Multi-Layer Perceptron, and XGBoost. Accuracy ranged from **74–79%**, but a critical flaw emerged: a **3.2-to-1 class imbalance** caused a *lazy model* problem in which classifiers inflated accuracy by predominantly predicting Non-Clickbait, yielding poor recall on the minority clickbait class.

3.3.2 Phase 2 — Environment Setup

Data loading was ported from a local Windows environment to Google Colab. `FileNotFoundError` caused by Windows path separators was resolved by explicitly loading renamed files directly into the Colab runtime before instantiating any pipeline component.

3.3.3 Phase 3 — Rigorous Optimisation

Three targeted interventions corrected the baseline deficiencies:

(a) **Transformer Upgrade.** `all-MiniLM-L6-v2` was replaced with `paraphrase-mpnet-base-v2`, producing **768-d** vectors that capture finer semantic nuances—e.g. distinguishing “*study reveals*” (journalistic) from “*what scientists discovered*” (clickbait-framed)—that low-dimensional embeddings compress away.

(b) **Class Imbalance Correction.** The 3.2-to-1 imbalance was explicitly quantified and mitigated by injecting `class_weight=‘balanced’` into SVM and MLP, and setting `scale_pos_weight` in XGBoost, forcing all models to penalise misclassification of the minority clickbait class.

(c) **Hyperparameter Search (GridSearchCV).** A `PredefinedSplit` from the validation partition allowed `GridSearchCV` to test hundreds of configurations without leaking test information. The critical outcome: the SVM’s optimal kernel shifted from *linear* to **RBF**, exploiting the non-linear geometry of the 768-d embedding space.

3.3.4 Classifier Comparison — Phase 1 vs Phase 3

Table 5: Semantic Approach: Baseline vs. Optimised Performance

Classifier	Ph.1 Acc	Ph.3 Acc
SVM (linear → RBF)	74–79%	85%
MLP	74–79%	improved
XGBoost	74–79%	improved

The SVM with RBF kernel and balanced class weighting on 768-d MPNet embeddings is selected as the **final semantic model** ($\approx 85\%$ test accuracy, substantially improved clickbait recall).

4 Feature Fusion

Building upon the three independent approaches (lexical, syntactic, and semantic), we construct a **feature-level fusion model** by concatenating all feature representations into a unified vector space.

4.1 Fusion Strategy

Let:

$$X_{fusion} = [X_{lexical} \mid X_{syntactic} \mid X_{semantic}]$$

The final feature space consists of:

- Lexical features: $\sim 10,000$ dimensions (TF-IDF + char + handcrafted)
- Syntactic features: $\sim 3,096$ dimensions (POS + structural)
- Semantic features: 384 dimensions (MiniLM embeddings)

Total dimensionality: $\sim 13,480$ features.

This high-dimensional representation captures:

- Surface-level lexical cues (keywords, phrases)
- Grammatical structure (POS patterns)
- Deep semantic meaning (contextual embeddings)

4.2 Model Selection

Due to the sparse and high-dimensional nature of the fused feature space, **linear classifiers** remain the most effective.

Two models were evaluated:

- Logistic Regression (LR)
- Linear Support Vector Machine (SVM)

Hyperparameters were tuned using GridSearchCV with class balancing.

Table 6: Fusion Model Performance (Validation Set)

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.86	0.628	0.749	0.6836
Linear SVM	0.86	0.64	0.75	0.6894

4.3 Fusion Model Results

4.4 Key Observations

1. Feature fusion improves recall significantly. Compared to lexical+syntactic models (recall ~ 0.57), fusion increases recall to ~ 0.75 , indicating better detection of clickbait instances.

2. Semantic features drive the improvement. The inclusion of embeddings helps detect paraphrased clickbait that lexical features fail to capture.

3. Linear SVM is the best overall model. It achieves the highest F1-score (0.6894) due to:

- Robustness in high-dimensional spaces
- Better margin optimisation than Logistic Regression

4. Precision-recall trade-off is controlled. While precision slightly decreases, the gain in recall leads to higher F1-score, which is more critical for clickbait detection.

4.5 Why Fusion Works

The improvement arises because:

- Lexical features capture *explicit phrases*
- Syntactic features capture *writing patterns*
- Semantic features capture *meaning and paraphrases*

Individually, each feature type is incomplete. Together, they provide a **holistic representation**, reducing both false positives and false negatives.

5 Results

5.1 Lexical System — Final Performance

Table 7: Lexical System: Final Evaluation Across All Splits

Split	N	Acc	P	R	F1
Validation (2017)	4,386	86.7	70.2	79.7	74.6
Test (2017)	18,979	85.2	66.1	74.1	69.9
Test (2016)	2,992	80.2	75.3	33.8	46.6

5.2 Generalisation Analysis

Table 8: Performance Degradation Across Domains

Metric	Val $\rightarrow$ Test 17	Test 17 $\rightarrow$ Test 16
Acc drop	-1.5%	-5.0%
F1 drop	-4.7%	-23.3%
Recall drop	-5.6%	-40.3%
AUC drop	—	-5.7%

The in-distribution gap (Val $\rightarrow$ Test 2017) is modest. The cross-dataset gap is driven by **recall collapse** (79.7% $\rightarrow$ 33.8%); AUC remains reasonable at 0.838, indicating discriminative power is retained — the issue is threshold calibration, not feature quality.

5.3 Semantic System — Performance Summary

Table 9: Semantic System: Phase 1 vs. Phase 3 Test Accuracy

Configuration	Test Acc (%)
SVM (linear, untuned)	74-79
MLP (untuned)	74-79
XGBoost (untuned)	74-79
SVM (RBF, balanced, GS)	85

5.4 Cross-System Comparison (Lexical vs. Syntactic vs. Semantic)

Table 10: All Approaches: 2017 In-Distribution Test Performance

Approach	Acc	F1
Lexical(LR + TF-IDF + Char + HF)	0.852	0.699
Syntactic(SVM + POS-TF-IDF)	0.800	0.810
Semantic (SVM-RBF + MPNet 768-d)	0.850	0.840
Fused	0.804	0.806

5.5 Overall Model Comparison and Best Performer

Across all approaches, the **semantic model (SVM with RBF kernel on MPNet embeddings)** emerges as the best-performing system, achieving approximately **0.85 F1-score** on the 2017 test dataset.

While the lexical and syntactic approaches capture important aspects of clickbait, they remain limited

Table 11: All Approaches: 2016 Cross-Dataset Performance

Approach	Final Model	Acc	F1
Lexical	LR + TF-IDF (1,3) + Char + HF	0.802	0.466
Syntactic	SVM + Handcrafted + POS-TF-IDF	0.790	0.770
Fused	0.805	0.820	

to surface-level patterns and structural cues. In contrast, the semantic model leverages deep contextual representations, allowing it to generalise beyond explicit keywords and fixed grammatical structures.

Comparison across approaches:

- **Lexical model:** Strong precision due to explicit phrase detection, but suffers from vocabulary dependence and reduced generalisation.
- **Syntactic model:** Captures structural patterns effectively and generalises well across datasets, but lacks semantic understanding.
- **Semantic model:** Achieves the best overall performance by capturing contextual meaning and paraphrased clickbait patterns.

Why the semantic model performs best:

The superior performance of the semantic approach can be attributed to three key factors:

- **Contextual understanding:** Transformer-based embeddings encode sentence-level meaning, allowing detection of clickbait even when explicit trigger words are absent.
- **Paraphrase robustness:** Models like MPNet are trained on paraphrase tasks, enabling recognition of semantically equivalent clickbait phrasing.
- **Non-linear decision boundary:** The RBF kernel SVM effectively separates complex patterns in the embedding space, unlike linear models used in other approaches.

Implications:

This result suggests that clickbait detection is fundamentally a **semantic problem** rather than purely lexical or syntactic. While feature engineering approaches provide interpretability and useful signals, deep contextual representations are more effective in capturing the underlying intent of headlines.

However, it is important to note that semantic models are less interpretable compared to lexical approaches, and their performance depends heavily on pre-trained embeddings.

6 Discussion

6.1 Syntactic Approach — Key Findings

Vocabulary-independence is a genuine strength. By operating purely on POS sequences, the syntactic model is immune to the vocabulary drift that collapses lexical recall on the 2016 corpus. Cross-dataset F1 drops only from 0.81 to 0.77 (combined model)—far shallower than the lexical model’s −23.3% F1 degradation.

POS n-grams generalise better than count features. Model 2 (POS-TF-IDF) consistently outperforms Model 1 (handcrafted counts) cross-dataset (F1: 0.79 vs. 0.76), because sequential patterns are more abstract than per-tag counts which can indirectly reflect vocabulary habits.

Combining features offers diminishing returns cross-dataset. Model 3 achieves the best in-distribution accuracy (0.80) but lower 2016 F1 than Model 2 alone (0.77 vs. 0.79), suggesting handcrafted count features introduce mild domain-specific bias that POS-TF-IDF alone avoids.

6.2 Semantic Approach — Key Findings

Why the RBF kernel mattered. Sentence Transformer embeddings occupy a high-dimensional, non-linear manifold. A linear SVM cannot separate these clusters efficiently, whereas the RBF kernel implicitly maps to an infinite-dimensional feature space, recovering ∼10 percentage points in accuracy over the linear baseline.

Class imbalance was the dominant failure mode. The 3.2-to-1 imbalance caused baseline models to exploit the majority class. Adding `class_weight=‘balanced’` without any other change substantially improved clickbait recall, confirming that the embedding quality was adequate but the optimisation objective was misaligned.

Encoder capacity matters. The 384-d MiniLM model trades semantic richness for speed. Upgrading to 768-d MPNet embeddings preserved contextual nuances—paraphrase-aware training means the encoder recognises that “*won’t believe*” and “*you must see*” are semantically equivalent clickbait patterns, even if surface forms differ.

6.3 Lexical Approach — Strengths & Limitations

Strengths:

- Strong in-distribution performance (85.2% accuracy, AUC 0.895) competitive with published classical baselines.

- Full interpretability via LR coefficients; linguistically coherent clickbait and non-clickbait profiles confirmed (Table ??).

- Fast inference after 46.7% feature-space reduction via χ^2 selection.

Limitations:

- *Vocabulary dependence:* Fixed-vocabulary models fail as clickbait language evolves to adopt news-style framing.
- *Threshold miscalibration:* Threshold $\theta=0.60$ (calibrated for 2017 data) is too conservative for the 2016 distribution, pushing borderline cases ($p \in [0.40, 0.59]$) to the non-clickbait class.
- *False Positives:* Legitimate journalism adopts audience-friendly framing (e.g., “*What You Need to Know About...*”) that overlaps with clickbait surface features.
- *False Negatives:* Sophisticated clickbait mimics journalistic form (e.g., “*New Study Reveals...*”), which lexical models cannot distinguish from genuine reporting.

These limitations motivate the syntactic and semantic approaches, which aim to capture higher-level structure and meaning *independent* of surface form.

6.4 Error Analysis — Confusion Matrices

Table 12: Confusion Matrix — Validation Set (2017)

		Predicted	
		Non-CB	Clickbait
Actual	Non-CB	1,890	299 (13.7% FPR)
	Clickbait	418	1,779 (81.0% TPR)

Table 13: Confusion Matrix — 2016 Cross-Dataset Test

		Predicted	
		Non-CB	Clickbait
Actual	Non-CB	1,383	117 (7.8% FPR)
	Clickbait	978	514 (34.5% TPR)

7 Conclusion

The lexical approach achieves strong in-distribution results through systematic multi-level feature engineering. Starting from a 63.0% F1 baseline, eleven progressive experiments yield a +11.6% improvement to **74.6% validation F1** (69.9% F1, AUC 0.895 on the held-out 2017 test set).

Core contributions:

- A 15,019-feature space (word TF-IDF (1, 3) + char n-grams (3, 5) + 19 domain features), reduced to 8,000 via χ^2 selection without performance loss.
- Full interpretability via LR coefficients, revealing linguistically meaningful clickbait signatures.
- Clear identification of cross-dataset failure modes: threshold miscalibration and vocabulary drift, both addressable through domain-adaptive recalibration.

The semantic approach demonstrates that transformer-based sentence embeddings provide a strong alternative to hand-crafted feature engineering. The Phase 3 optimised pipeline—`paraphrase-mpnet-base-v2` encoder, RBF-kernel SVM, balanced class weighting, and GridSearchCV tuning—achieves $\approx 85\%$ test accuracy, matching the lexical system’s in-distribution performance via an entirely different representational paradigm.

Key semantic contributions:

- Identification and correction of a 3.2-to-1 class imbalance that masked genuine model capability in the baseline.
- Encoder upgrade from 384-d to 768-d producing richer semantic representations.
- GridSearchCV with predefined validation split surfacing RBF as the optimal kernel, yielding a $\sim 10\%$ accuracy jump over the linear baseline.

The syntactic approach demonstrates that structure alone—without any access to word identity—is sufficient to achieve competitive performance (0.80 accuracy, 0.81 F1 in-distribution). Three syntactic models were developed:

- *Handcrafted features* (M1): interpretable POS counts and structural cues; 0.79 accuracy, 0.80 F1 in-distribution.
- *POS-TF-IDF* (M2): sequential n-gram patterns; best cross-dataset generalisation (2016 F1: 0.79).
- *Combined* (M3): best in-distribution accuracy (0.80), with the notable insight that POS n-grams alone generalise better cross-dataset than hand-crafted counts.

[Conclusions for the syntactic and fused systems will be appended upon completion. A unified finding on which approach best balances accuracy, generalisability, and interpretability will follow from Table 10.]

8 How could the clickbait detection system be improved? Why are these improvements feasible?

1. Fine-tuning transformer models. Instead of using fixed sentence embeddings, fine-tuning a transformer directly on the clickbait dataset can im-

prove performance by learning task-specific patterns such as misleading phrasing and curiosity gaps. This is feasible since modern transformer models are designed for downstream fine-tuning and have consistently shown improvements over static embeddings.

2. Better handling of class imbalance. Although class weights are used, techniques like focal loss or resampling can further improve recall for clickbait instances. This is possible because the imbalance in the dataset is already known and can be directly addressed during training.

3. Incorporating additional context. Using only headlines can limit understanding. Including article content or metadata can provide richer context and improve classification accuracy. This is feasible as such information is typically available alongside headlines in real-world datasets.